

Solving the Reproducibility Crisis

MedInsure Case Study: Issue #5 – Ensuring Trust & Auditability in AI

The Crisis: Non-Reproducible Models

The Failure

MedInsure's Data Science team successfully built a claim approval model but failed the pre-production audit.

When asked to prove the exact training conditions, the team could not reproduce the run.

The Risk

Without reproducibility, the model is a "black box." We cannot verify its fairness, accuracy, or lineage, making it ineligible for production deployment in a regulated industry.



Why Did We Fail the Audit?



Untracked Data

The historical claim data used for training was not versioned. We cannot prove which specific records influenced the model.



Environment Drift

Training happened on Python 3.8, but staging used 3.10. Library versions weren't locked, causing prediction mismatches.



Missing Metadata

Critical settings like hyperparameters and random seeds were not logged, preventing exact recreation of the model state.



The MLOps Solution

Implementing the Three Pillars of
Reproducibility

Pillars 1 & 2: Data and Environment

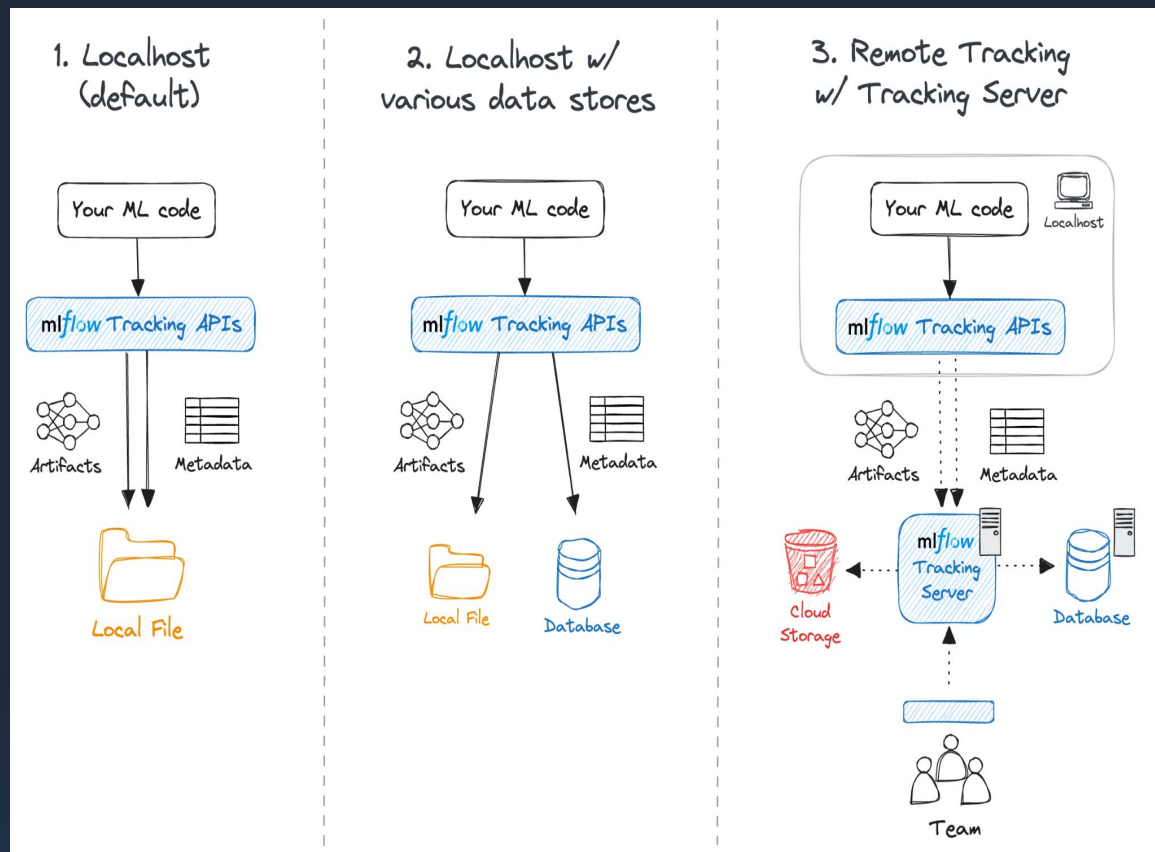
1. Data Version Control

We must treat data like code. By using tools like **DVC**, we create a snapshot of the exact 5-year claim history used for training. This assigns a unique ID (e.g., `claims_v3.2`) to the dataset, ensuring we can always retrieve the correct training input.

2. Containerization

To solve the runtime mismatch, we wrap the model in a **Docker Container**. This freezes the operating system, Python version (3.8), and all library dependencies. The model now runs in an identical environment, whether on a laptop or the production server.

Pillar 3: Experiment Tracking



Automated Governance

We need to stop relying on manual notes. By integrating an Experiment Tracking tool (like MLflow), we automatically capture:

- **Hyperparameters:** Learning rates, epoch counts.
- **Metrics:** Precision, Recall, Accuracy.
- **Code Version:** The exact Git commit hash.

This creates an unbreakable link between the code, the config, and the result.

The Result: Full Auditability

By implementing these pillars, we establish a **Model Registry** that acts as a single source of truth.

Approved Model Record:

- Establishment of single source of truth
- Full auditability
- verifiable model record

This provides the irrefutable evidence required by auditors to approve production deployment.

